Expt. 26 Millikan oil-drop experiment

Ⅰ. Purpose

Master the method to measure the value of the electron charge e and illustrate that all electric charges are integral multiples of the elementary charge.

Ⅱ. Principle

Use an atomizer to spray oil mist between two parallel plates. Due to friction, the oil drops formed are generally charged. When there is no voltage between the plates the oil drops will fall under the action of gravity. There is accelerated motion at first followed by deceleration under the action of the viscous force of the air in which the oil drops are falling. When the viscous force F_r, the buoyancy force F_f and the gravity mg are in equilibrium, the oil drop will fall with uniform velocity v_g, given by

$$mg - F_f - F_r = 0 \tag{26-1}$$

Because of the action of surface tension, the oil drop is spherically shaped. Assuming that the oil drop has a radius a and density of ρ, then the mass m is given by $m = \frac{4\pi}{3}a^3\rho$. If the air density is ρ', then the buoyancy force of the air on the oil drop is $F_f = \frac{4\pi}{3}a^3\rho' g$. According to Stokes' law, the viscous force on the oil drop is $F_r = 6\pi\eta a v_g$, where η is the viscosity coefficient of the air. Equation (26-1) can now be written

$$\frac{4\pi}{3}a^3\rho g - \frac{4\pi}{3}a^3\rho' g - 6\pi\eta a v_g = 0 \tag{26-2}$$

Since $\rho' \ll \rho$, equation (26-2) gives

$$a \approx \sqrt{\frac{9\eta v_g}{2g\rho}} \tag{26-3}$$

When Millikan conducted his experiment, he mounted big parallel electrodes plate both above and below the space in which the oil drop moved, as shown as Figure 26-1. When there is voltage U between the parallel plates it generates a uniform electric field between the plates, $E = U/d$, where d is the distance between the plates. Here the oil drop with electric charge q suspended between the plates will be under the action of an electric field force qE. If the electric field force

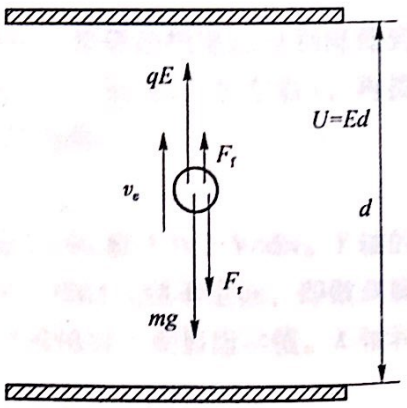


Figure 26-1 Force Analysis on a Charged Oil Drop in an Electric Field

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$qE > mg$, the oil drop will move upward provided the two forces are in opposite directions. The magnitude and direction of the viscous force will depend upon the magnitude and direction of the velocity of the oil drop. When the velocity of the oil drop increases to v_e, the resultant force again achieves equilibrium. The oil drop will move upward at a uniform velocity v_e and the viscous force on the oil drop is $F_r = 6\pi\eta a v_e$, so we can now write

$$mg + F_r - F_f - qE = 0 \tag{26-4}$$

i. e.

$$\frac{4\pi}{3}a^3\rho g + 6\pi\eta a v_e - \frac{4\pi}{3}a^3\rho' g - qE = 0 \tag{26-5}$$

As before since $\rho' \ll \rho$, we can ignore the influence of the air buoyancy force, so the electrical charge q on the oil drop is given by

$$q = 18\pi \frac{d}{U}\left(\frac{\eta^3}{2\rho g}\right)^{\frac{1}{2}} v_g^{\frac{1}{2}}(v_g + v_e) \tag{26-6}$$

ρ and η can be obtained by referring to the table. So if we measure the uniform motion velocities v_g and v_e of the oil drop between the two plates we can find out the charge q (units in coulombs) through using equation (26-6).

Millikan found that when the oil drop radius approaches the magnitude of the mean free path λ of the air, the air medium cannot be considered as continuous and Stokes' law should be modified, i. e. :

$$F_r = \frac{6\pi a\eta v}{1 + \frac{b}{pa}} \tag{26-7}$$

Where $b = 6.17 \times 10^{-6}$ (m · cm mercury column) $= 8.22 \times 10^{-3}$ (Pa · m) is called the correction constant. $p = 76.0\text{cmHg} = 1.013 \times 10^5\text{Pa}$ is the air pressure. The equation (26-3) can be modified as

$$a = \sqrt{\frac{9\eta v_g}{2\rho g\left(1 + \frac{b}{pa}\right)}} \tag{26-8}$$

Notes: the oil drop radius a is included in the square root sign of the above expression, but since it is in the correction term, with only a small influence, when a is calculated approximately, the influence of this term can be ignored.

During the experiment, you should use the same distance for the oil drop falling and rising at uniform velocity. Assume that this distance is L, and take the falling time and rising time of the oil drop as t_g and t_e respectively, then we have

$$v_g = L/t_g, \quad v_e = L/t_e \tag{26-9}$$

equation (26-6) can now be written as:

$$q = K\left(\frac{1}{t_g} + \frac{1}{t_e}\right)\left(\frac{1}{t_g}\right)^{\frac{1}{2}}/U \tag{26-10}$$

where

$$K = \frac{18\pi}{\sqrt{2\rho g}}\left(\frac{\eta L}{1 + \frac{b}{pa}}\right)^{\frac{3}{2}} \cdot d$$

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expression (26 – 10) is the principal formula for measuring the electric charge on the oil drop by the Dynamic (Non-equilibrium) Method.

In the above formulae,

Density of oil $\rho = 981\ \text{kg} \cdot \text{m}^{-3}$ (20°C);

Acceleration of gravity $g = 9.801\text{m} \cdot \text{s}^{-2}$ (Peking);

Coefficient of viscosity of air $\eta = 1.83 \times 10^{-5}\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1} = 1.83 \times 10^{-5}\text{Pa} \cdot \text{s}$;

Uniform velocity distance $L = 1.5 \times 10^{-3}\text{m}$;

Distance between parallel plates $d = 6.00 \times 10^{-3}\text{m}$.

Millikan also made measurements on the oil drop electric charge by a Static (Equilibrium) Method. The method is to adjust the voltage between the parallel plates, bringing the oil drop to rest i. e. $v_e = 0$, $t_e \to \infty$, Then from equation (26 – 10) we have that

$$q = K\left(\frac{1}{t_g}\right)^{\frac{3}{2}}/U \qquad (26 - 11)$$

Measurements on the same oil drop using both methods should give the same results.

In the experiment, you select different oil drops to measure. With the electric charge expressed as q_1, q_2, ..., q_i, after analysis it can be found that these quantities are all of integral multiple of the elementary charge e, i. e.

$$q_i = n_i \times e \qquad n_i = \pm 1, \pm 2, \pm 3, \ldots \qquad (26 - 12)$$

This illustrates the existence of the minimum electric charge unit-the electron charge e.

Ⅲ. Experimental

1. Practice measurement

To know functions of the keys on the panel of the instrument well, you need to practice to control the oil drop motion. You need to master the method of measurement of the oil drop motion time and learn to select appropriate oil drops.

As shown in Figure 26 – 2, there are only two switches to control the parallel plate electrode voltage on the panel: switch K_1 controls the polarity of the upper plate voltage; switch K_2 controls the magnitude of the upper plate voltage. When K_2 is in the mid position, i. e. "Switch" position, it can utilize the equilibrium voltage adjusting potentiometer to control the magnitude of the equilibrium voltage. At this moment, when the switch is turned to the "Raising" position, the instrument automatically adds a raising voltage of 200 – 300V to the equilibrium voltage. When it is turned to the "0V" position, the voltage on the upper plate is 0V.

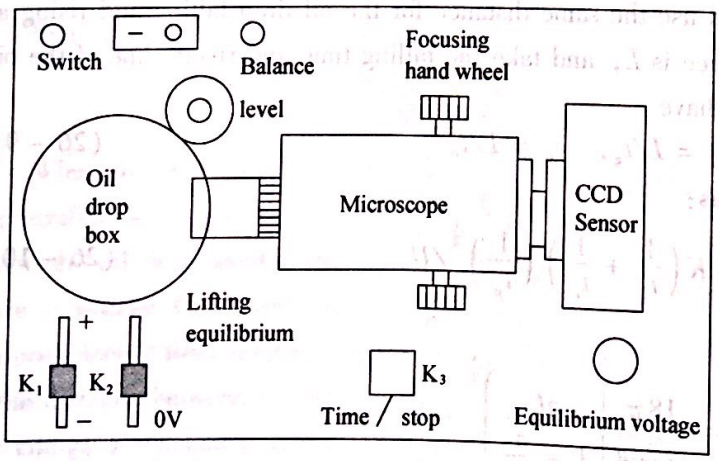


Figure 26 – 2 Diagram of the Oil Drop Measuring Instrument Panel

When Switch K_2 is turned from

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"Equilibrium" to the "0V" position, the oil drop starts to accelerate downwards and the instrument will automatically starts counting time. When the oil drop falls to the expected position, quickly turn the switch back from "0V" to the "Equilibrium" position and the oil drop will stop moving and the timer stops counting. The time displayed on the upper right corner of the monitor screen at this moment is the time corresponding to the oil drop motion.

Because of the existence of air resistance, the time for the oil drop's accelerated motion is very short and it can be considered that the oil drop begins uniform motion instantaneously. Suddenly adding the equilibrium voltage on the moving oil drop, causes it to immediately come to rest.

The "Timing/Stop" key of the oil drop measuring instrument-Switch K_3. Press once to set the timer to 0 and start the time counting, press again to stop counting and save the data. This switch is used in measuring t by the dynamic method.

It is very important to select an appropriate oil drop. Big and bright oil drops are certainly of greater mass, as well as more electric charge, causing a very short time for falling at uniform velocity and adding uncertainty for time measurements. Generally it is appropriate to select oil drops with an equilibrium voltage of about 200V (both " + " and " − " are available), 1.5 mm falling distance at uniform velocity, time spent about 8 − 20 s (total height of the 8 grids is 2 mm). After spraying oil, turn Switch K_2 to the "equilibrium" position. Adjust the equilibrium voltage to 200 − 300V and notice several slowly moving, clear and bright oil drops. Try to turn Switch K_2 to "0V" position, observing the approximate velocity of the oil drops, selecting one from them as the measuring object. For the 9 inch monitor, oil drops with diameter about 0.5 − 1 mm in size are appropriate.

Determining whether the oil drop is in equilibrium requires much patience. Move the oil drop to a certain calibration line through use of the control switches (in the experiment use the second line above the oil drop start line). Carefully adjust the equilibrium voltage, repeating several times, and only after you observe for a period of time to find that the oil drop definitely does not move any more you can consider that it is force equilibrium.

Note: to measure the time needed for the oil drop rising or falling a certain distance, you should choose particular calibration lines to be the timing reference points for the oil drop. Measure the same oil drop to practice repeatedly in order to reduce the variation in the measured time values.

2. Formal measurement

Adopt the "equilibrium" method for your measurements. Move the equilibrium adjusted oil drop to the "Start" line by using Switch K_2. Press the "Timing/Stop" key, making the timer stop. Then turn the Switch K_2 to "0V" position. At that time that the oil drop starts to fall and the timer starts counting. When the oil drop reaches the "Finish" line (the next to last traversed line), quickly turn Switch K_2 to the "equilibrium" position. The oil drop stops moving immediately, and the timer stops timing. The time counted is the time t_g for the oil drop uniform motion. The moving distance for the oil drop is $L = 0.25 \times 6 = 1.5\text{mm}$. Thereby you can determine the falling velocity v_g of the oil drop. Taking this result into equition (26 − 11), you can find the electric charge q of this oil drop.

The "Dynamic" method requires you to separately measure the oil drop rising time t_e with rising

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voltage on the upper plate and the oil drop falling time t_g with no voltage on the upper plate. Take the measured result into equation (26 - 10), so you can determine the electric charge q of the measured oil drop.

This experiment requires you to measure the selected oil drop 7 times. Select 5 oil drops (equilibrium method) and 2 oil drops (dynamic method) to measure. Find out the magnitudes of electric charge q and electron charge e of for each oil drop. Compare the measured value with the accepted value ($e_0 = 1.60217733 \times 10^{-19}$ coulomb). Determine the relative uncertainty of your measured result.

Ⅳ. Data Analysis

1. Static method

Utilize the microcomputer data processing software for this experiment to conduct a self check on the measured results of the static method. For all measured data of uncertainty over 5%, you will have to analyze the reason and repeat the measurement. You have to select 5 independent oil drops to complete the experiment, and finish the data analysis.

2. Dynamic method

Select 2 oil drops, measuring 7 times for each drop. Separately design the data analysis table for the dynamic measuring method.

The data analysis of the dynamic method needs to be proved. In order to simplify it, you can identify the value of K in equation (26 - 11) and hence calculate the value of a.

Notes:

(1) Oil in the atomizer should not be overfilled.

(2) Every time, at the end of the experiment, you should wipe off the residual oil on the electrode plates using a clean paper towel. In addition take off the air cell of the atomizer, to avoid damage by spilled oil.

(3) Before wiping the atomizer, you should put switch K_1 and K_2 to the position of zero voltage and turn off the power switch of the instrument.

Ⅴ. Questions

1. What is the reason for the oil drop motion trajectory not being perpendicular to the ground in the experiment?

2. What are the relationships and differences between the definitions of voltage U in equation (26 - 10) and equation (26 - 11)?

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实验二十六　密立根油滴实验

一、实验目的

掌握测量电子电荷 e 值的方法，并证明所有电荷都是基本电荷 e 的整数倍。

二、实验原理

用喷雾器将油雾滴喷入两块平行的电极板之间，油滴在喷射时由于摩擦，一般都是带电的。当两极板未加电压时，悬浮在空气中的油滴受到重力作用而降落，开始时做加速运动，但随即受到空气黏滞力的作用而减速。当油滴受到的空气黏滞力 F_r 和浮力 F_f 与重力 mg 的合力平衡时，则油滴将以速度 v_g 匀速下降，此时有

$$mg - F_f - F_r = 0 \tag{26-1}$$

由于表面张力的作用油滴呈球状。设油滴的半径为 a，密度为 ρ，则 $m = \frac{4\pi}{3}a^3\rho$。如果空气密度为 ρ'，则空气对油滴的浮力为 $F_f = \frac{4\pi}{3}a^3\rho' g$。根据斯托克斯定律，油滴所受黏滞力 $F_r = 6\pi\eta a v_g$，其中 η 是空气的黏滞系数。则式（26-1）可以写成

$$\frac{4\pi}{3}a^3\rho g - \frac{4\pi}{3}a^3\rho' g - 6\pi\eta a v_g = 0 \tag{26-2}$$

考虑到 $\rho' \ll \rho$，由式（26-2）可得

$$a \approx \sqrt{\frac{9\eta v_g}{2g\rho}} \tag{26-3}$$

密立根在进行此实验时，在油滴运动空间的上、下方各安装了一个大的平行电极板，如图 26-1 所示。当两平行极板间加有电压 U 时，两板之间的中心区域将产生均匀电场 $E = U/d$，d 为极板间距离。悬浮在极板间带有电荷 q 的油滴此时将受到电场力 qE 的作用。如果，电场力 $qE > mg$，而且两力的方向相反，油滴就会向上运动。此时黏滞力的大小、方向也将随油滴运动速度大小与方向的改变而变化。当油滴速度增大到 v_e 时，所受合力又达平衡，油滴就将以速度 v_e 匀速上升，此时油滴所受黏滞力为 $F_r = 6\pi\eta a v_e$，故有下式

$$mg + F_r - F_f - qE = 0 \tag{26-4}$$

即

$$\frac{4\pi}{3}a^3\rho g + 6\pi\eta a v_e - \frac{4\pi}{3}a^3\rho' g - qE = 0 \tag{26-5}$$

同理忽略空气浮力的影响，可得油滴所带电量为

$$q = 18\pi \frac{d}{U}\left(\frac{\eta^3}{2\rho g}\right)^{\frac{1}{2}} v_g^{\frac{1}{2}}(v_g + v_e) \tag{26-6}$$

式中 ρ 和 η 可查表获得。所以只要测出油滴在两极板之间匀速运动时的 v_g 和 v_e，便可由式

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(26－6）求出油滴所带电量 q（单位是库仑）。

密里根发现，当油滴半径 a 与空气的平均自由程 λ 量级接近时，空气介质不能认为是连续的，斯托克斯定律应当进行修正，即：

$$F_r = \frac{6\pi a\eta v}{1+\frac{b}{pa}} \tag{26-7}$$

式中，$b=6.17^{-6}$（m·cmHg）$=8.22\times10^{-3}$（Pa·m）为修正常数。

$p=76.0\text{cmHg}=1.013\times10^5\text{Pa}$（大气压强），于是，式（26－3）应修正为

$$a=\sqrt{\frac{9\eta v_g}{2\rho g\left(1+\frac{b}{pa}\right)}} \tag{26-8}$$

注意：式（26－8）中根号内仍包含油滴半径 a，但由于它处于修正项中，影响较小，在 a 的近似计算时，可忽略此项的影响。

实验时取油滴匀速下降和匀速上升的距离相同，设为 L，测出油滴下降与上升时间分别为 t_g 与 t_e，则

$$v_g=L/t_g,\quad v_e=L/t_e \tag{26-9}$$

式（26－6）可表述为：

$$q=K\left(\frac{1}{t_g}+\frac{1}{t_e}\right)\left(\frac{1}{t_g}\right)^{\frac{1}{2}}/U \tag{26-10}$$

式中，

$$K=\frac{18\pi}{\sqrt{2\rho g}}\left(\frac{\eta L}{1+\frac{b}{pa}}\right)^{\frac{3}{2}}\cdot d$$

式（26－10）即为动态（非平衡）法测定油滴带电量的原理公式。

式中：油的密度　　$\rho=981\text{kg}\cdot\text{m}^{-3}$　　（20°C）；

重力加速度　　$g=9.801\text{m}\cdot\text{s}^{-2}$　　（北京）；

空气黏滞系数　　$\eta=1.83\times10^{-5}\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}=1.83\times10^{-5}\text{Pa}\cdot\text{s}$；

油滴匀速下降距离　　$L=1.5\times10^{-3}\text{m}$；

平行极板间距离　　$d=6.00\times10^{-3}\text{m}$。

密里根也曾利用静态（平衡）法完成油滴带电量的测量。其原理是：调节平行极板间的电压，使油滴处于静止状态，即 $v_e=0$，$t_e\to\infty$，则由式（26－10）可得

$$q=K\left(\frac{1}{t_g}\right)^{\frac{3}{2}}/U \tag{26-11}$$

对同一个油滴采用不同的方法测量，所得的结果应是一致的。

在实验中选择不同的油滴进行测量，其带电量可以表示为 q_1，q_2，…，q_i，经过分析可发现，这些量都是基本电荷电量 e 的整倍数，即

$$q_i=n_i\times e \qquad n_i=\pm1,\pm2,\pm3,\cdots \tag{26-12}$$

这就证明了存在着最小的电荷单位——电子电荷 e。

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三、实验内容与步骤

1. 测量练习

熟悉仪器面板上各键的功能、练习控制油滴运动、掌握油滴运动时间的测量方法、学会选择合适的油滴。

图 26－2 所示的箱体面板上有两只控制平行极板电压的电键：K_1 键控制上极板电压的极性；K_2 键控制上极板电压的大小。当 K_2 键处于中间位置，即“平衡”挡时，可用平衡电压调节电位器控制平衡电压的高低。当此电键打向“提升”挡时，本机自动在该平衡电压的基础上增加 200～300V 的提升电压。而打向“0V”挡时，上极板上的电压为 0V。

在 K_2 键由“平衡”打向“0V”时，油滴开始加速下落的同时本机开始自动计时，油滴下落到预定位置时，迅速将该键由“0V”挡扳回“平衡”挡，油滴将停止运动，与此同时计时器停止计时。此时监视屏右上端显示的时间就是油滴运动对应的时间。

由于空气阻力的存在，油滴做加速运动的时间非常短，可认为油滴在瞬间即开始做匀速运动。运动的油滴突然加上平衡电压，将立即停止运动而静止下来。

油滴仪的“计时/停”按键——K_3 键，按一下开关，将在清 0 的同时开始计数，再按一下，停止计数，并保存数据。该键应用于动态法 t_e 的测量。

选择一颗合适的油滴十分重要。大而亮的油滴必然质量大，所带电荷也多，而匀速下降的时间则很短，增大了测时的不确定度。通常，选择平衡电压为 200V 左右（“＋”或“－”均可），匀速下落 1.5mm 的距离，所用时间在 8～20s 之间的油滴较合适（8 格的总高度为 2mm）。喷油后，将 K_2 键置于“平衡”挡，调节平衡电压为 200～300V，注意几颗缓慢运动、较为清晰明亮的油滴。试将 K_2 键置于“0V”挡，观察各颗油滴下落的大概速度，从中选一颗为测量对象。对于 9in 的监视器，目视直径在 0.5～1mm 之间的油滴较合适。

判断油滴是否平衡要有足够的耐性。用控制键将油滴移至某条刻度线上（实验中油滴起点在上方第二条线处），仔细调节平衡电压，这样反复操作几次，经过一段时间观察油滴确实不再移动才能认为其受力平衡了。

注意：测量油滴上升或下降某段距离所需的时间，应该统一油滴到达刻度线什么位置作为计时的参考点。利用同一颗油滴反复多练习几次，使测出的各次时间的离散性较小。

2. 正式测量

采用“平衡”法测量。可将已调平衡的油滴用 K_2 键移到“起点”线上。按“计时/停”按钮，使计时器停止计时，然后，将 K_2 键拨向“0V”挡，油滴开始下降的同时，计时器自动开始计时。油滴到达“终点”（倒数第二条横线）时，迅速将 K_2 键拨向“平衡”挡，油滴可立刻停止运动，计时器也即刻停止计时。其所计时间为油滴匀速运动所用时间 t_g。油滴运动距离 $L=0.25\times6=1.5\text{mm}$。由此可确定该油滴的下落速度 v_g。将此结果代入式（26－11），即可求出该油滴所带电量 q 值。

“动态”法要求分别测出上极板加提升电压时油滴的上升时间 t_e 和上极板不加电压时油滴下落的时间 t_g，而将测量结果代入式（26－10），即可确定被测油滴所带电量 q 值。

本实验要求：对选定的油滴反复测量 7 次。选择 5 颗油滴（平衡法）、2 颗油滴（动态法）分别进行测量。求出每颗油滴的带电量 q 及电子电荷 e 的量值。并将结果与公认值

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（$e_0 = 1.60217733 \times 10^{-19}$库仑）进行比较，求出测量结果的相对不确定度值。

四、数据处理

1. 静态法

利用本实验的微机数据处理软件对静态法的测量结果进行自检。凡相对不确定度超过5%的测量数据需要分析原因，并重新测量。需选择5个独立的油滴完成实验，完成各个油滴的数据处理。

2. 动态法

选择2颗油滴，每颗油滴各测量7次。自行设计动态法测量的数据处理表格。

动态法的数据处理需要有证明过程。为了简化起见，可将式（26－11）中的K值先确定下来，并计算a值。

注意：

（1）喷雾器内的油不可装得太满，防止溢出。

（2）每次实验完毕，应及时用纸巾擦净上电极板及油雾室内的积油，并将喷雾器的气囊取下，防止溢油浸坏气囊。

（3）擦拭油雾杯前，应将K_1、K_2电键放置在电压为零的位置并关闭仪器的电源开关。

五、思考题

1. 在实验中油滴运动轨迹不垂直地面的原因是什么？
2. 公式（26－10）和公式（26－11）中的电压U的定义有何联系与区别？

由 扫描全能王 扫描创建

实验二十一　密立根油滴实验

1. 平衡法(某一电荷 q_i)

序　数	U/V	t_g/s	V_g/m·s^{-1}	q_i/c	n_i(个)	e/c×10^{-19}	u_e/e_0
1							
2							
3							
4							
5							
平均值							

2. 动态法

参看上表,自行设计动态法测量的数据表格。

思考题

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